

ALL ASSIGNMENTS [QUIZZES AND PROGRAMMING]

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Week 1 Quiz

The due date for submitting this assignment has passed.

Due on 2025-02-05, 23:59 IST.

Score: 10/10=100%

Assignment submitted on 2025-02-02, 07:52 IST

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

1) An algorithm has two phases. The first phase, initialization, takes time $O(n^3)$. The second phase, which is the main computation, takes time $O(n^2)$. What is the most accurate description of the complexity of the overall algorithm?

2 points

- ☐ $O(n^6)$
- ☐ $O(n^5)$
- ☒ $O(n^3)$
- ☐ $O(n^2)$

Yes, the answer is correct.

Score: 2

Feedback:

When there are multiple phases in sequence, the largest of the phases dominates the overall complexity. Here the main computation is $O(n^2)$ and the initialization phase is $O(n^3)$, so the overall complexity is $O(n^3)$.

Accepted Answers:

$O(n^3)$

2) We are using a computer that performs 10^8 basic operations per second. We are trying to determine the worst case time complexity of a library function that is provided to us, whose code we cannot read. We test the function by feeding large numbers of random inputs of different sizes. We find that for inputs of size 50 the function always returns well within one second, for inputs of size 500 it sometimes takes a couple of seconds and for inputs of size 5,000 it sometimes takes over 15 minutes. What is a reasonable conclusion we can draw about the worst case time complexity of the library function?

2 points

- ☐ $O(n^4)$
- ☒ $O(n^3)$
- ☐ $O(n^2 \log n)$
- ☐ $O(n^2)$

Yes, the answer is correct.

Score: 2

Feedback:

Assuming a CPU with 10^8 operations per second, 50^2 would be well under 10^8 , whereas 500^2 is 1.25×10^8 and 5000^2 is 1250×10^8 (about 21 minutes). Note that 5000^2 is 0.25×10^8 and $\log 5000$ is about 13, so if the complexity were $O(n^2 \log n)$, the worst case running time for an input of size 5000 would be a few seconds, not 15 minutes.

Accepted Answers:

$O(n^2)$

3) Suppose $f(n)$ is $2n^2+4n+5$ and $g(n)$ is $7n^3 + 5n^2 + 12$. Let $h(n)$ be a third, unknown function. Which of the following is **not** possible.

2 points

- ☐ $h(n)$ is $O(f(n))$ and $h(n)$ is also $O(g(n))$
- ☐ $h(n)$ is not $O(f(n))$ and $h(n)$ is also not $O(g(n))$
- ☒ $h(n)$ is $O(f(n))$ but $h(n)$ is not $O(g(n))$
- ☐ $h(n)$ is $O(g(n))$ but $h(n)$ is not $O(f(n))$

Yes, the answer is correct.

Score: 2

Feedback:

Since $f(n)$ is $O(g(n))$, if $h(n)$ is $O(f(n))$ it must also be $O(g(n))$. All other combinations are possible.

Accepted Answers:

$h(n)$ is $O(f(n))$ but $h(n)$ is not $O(g(n))$

4) How many times is the comparison $i <= n$ performed in the following program?

2 points

```
int i = 60, n = 300;
main(){
    while (i <= n){
        i = i+2;
        n = n-3;
    }
}
```

- ☐ 48
- ☐ 49
- ☒ 50
- ☐ 51

Yes, the answer is correct.

Score: 2

Feedback:

After 48 iterations, i is $60+96=156$ and n is $300-144=156$. At this point, $i==n$ so the 49th iteration begins. The next test of the while condition fails and exits the loop. Hence, overall the while condition is checked 50 times.

Accepted Answers:

50

5) If $T(n)$ is $O(n^{4/3})$ which of the following is **false**?

2 points

- ☒ $T(n)$ is $O(n \log n)$
- ☐ $T(n)$ is $O(n^2)$
- ☐ $T(n)$ is $O(n^2 \log n)$
- ☐ $T(n)$ is $O(n^3)$

Yes, the answer is correct.

Score: 2

Feedback:

$n^{1/3}$ is not $O(\log n)$

Accepted Answers:

$T(n)$ is $O(n \log n)$

Week 2 Quiz

The due date for submitting this assignment has passed.

Due on 2025-02-05, 23:59 IST.

Score: 8/10=80%

Assignment submitted on 2025-02-05, 22:29 IST

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

1) Having collected Aadhaar information for each SIM card, a government agency wants to check that all the Aadhaar numbers entered in SIM card applications are actually valid, by comparing SIM card applications against the list of registered Aadhaar numbers. Which of the following is likely to be the best strategy for this? **2 points**

- ☐ Use a nested loop. For each SIM card application S, and for each Aadhaar number A, check if the Aadhaar number listed in S is the same as A.
- ☐ Sort the SIM card applications and use binary search to speed up the process.
- ☒ Sort the registered Aadhaar numbers and use binary search to speed up the process.
- ☐ Sort both the SIM card applications and registered Aadhaar numbers and merge the two lists.

Yes, the answer is correct.

Score: 2

Feedback:

The nested loop is the correct naive strategy. Sorting the list in the inner loop allows us to use binary search instead of a nested loop, so the complexity reduces from $O(n^2)$ to $O(n \log n)$.

Accepted Answers:

Sort the registered Aadhaar numbers and use binary search to speed up the process.

2) Suppose our aim is to sort an array in descending order. Which of the following statements is true? **2 points**

- ☐ Input in ascending order is worst case for both selection sort and insertion sort.
- ☒ Input in ascending order is worst case for insertion sort but not for selection sort.
- ☐ Input in descending order is worst case for both selection sort and insertion sort.
- ☐ Input in ascending order is worst case for selection sort but not for insertion sort.

No, the answer is incorrect.

Score: 0

Feedback:

Input in ascending order and descending order are both worst case for selection sort. Input in ascending order is worst case for insertion sort.

Accepted Answers:

Input in ascending order is worst case for both selection sort and insertion sort.

3) Suppose we want to sort an array in ascending order. We have a stable implementation of quicksort that alternately chooses the first and last element of the array as the pivot with each recursive call. Which of the following is **not true** for this implementation? **2 points**

- ☒ The worst case behaviour is $O(n \log n)$.
- ☐ The average case behaviour is $O(n \log n)$.
- ☐ The worst case behaviour is $O(n^2)$.
- ☐ The average case behaviour is asymptotically better than the worst case behaviour.

Yes, the answer is correct.

Score: 2

Feedback:

For any fixed strategy to choose the pivot, we can reverse-engineer a worst case input that requires time $O(n^2)$.

Accepted Answers:

The worst case behaviour is $O(n \log n)$.

4) Which of the following statements is **not** true? **2 points**

- ☐ Quicksort and merge sort are both examples of divide and conquer algorithms.
- ☒ If we randomly choose a pivot element each time, quicksort will always terminate in time $O(n \log n)$.
- ☐ For every fixed strategy to choose a pivot for quicksort, we can construct a worst case input that requires time $O(n^2)$.
- ☐ If we could find the median in time $O(n)$, quicksort would have worst case complexity $O(n \log n)$.

Yes, the answer is correct.

Score: 2

Feedback:

Even with a random choice of pivot in each iteration, we can only guarantee that the expected time is $O(n \log n)$, not the worst-case.

Accepted Answers:

If we randomly choose a pivot element each time, quicksort will always terminate in time $O(n \log n)$.

5) We have a list of pairs [{"Shweta",71}, {"Sunita",85}, {"Tariq",71}, {"Brinda",85}, {"Salma",72}, {"Uday",60}], where each pair consists of a student's name and his/her marks in a course. We sort these pairs in ascending order of marks. Which of the following corresponds to a stable sort of this input? **2 points**

- ☐ [{"Uday",60}, {"Tariq",71}, {"Shweta",71}, {"Salma",72}, {"Sunita",85}, {"Brinda",85}]
- ☒ [{"Uday",60}, {"Shweta",71}, {"Tariq",71}, {"Salma",72}, {"Sunita",85}, {"Brinda",85}]
- ☐ [{"Uday",60}, {"Tariq",71}, {"Shweta",71}, {"Salma",72}, {"Brinda",85}, {"Sunita",85}]
- ☐ [{"Uday",60}, {"Shweta",71}, {"Tariq",71}, {"Salma",72}, {"Brinda",85}, {"Sunita",85}]

Yes, the answer is correct.

Score: 2

Feedback:

The repeated marks values are 71 and 85. The students with these marks should appear in the same order in the output as in the input.

Accepted Answers:

[{"Uday",60}, {"Shweta",71}, {"Tariq",71}, {"Salma",72}, {"Sunita",85}, {"Brinda",85}]

The Siruseri Singing Championship

(Zonal Computing Olympiad 2019)

The Siruseri Singing Championship is going to start, and Lavanya wants to figure out the outcome before the tournament even begins! Looking at past tournaments, she realizes that the judges care only about the pitches that the singers can sing in, and so she devises a method through which she can accurately predict the outcome of a match between any two singers.

She represents various pitches as integers and has assigned a lower limit and an upper limit for each singer, which corresponds to their vocal range. For any singer, the lower limit will always be less than the upper limit. If a singer has lower limit L and upper limit U ($L < U$), it means that this particular singer can sing in all the pitches between L and U , that is they can sing in the pitches $\{L, L+1, L+2, \dots, U\}$.

The lower bounds and upper bounds of all the singers are distinct. When two singers S_i and S_j with bounds (L_i, U_i) and (L_j, U_j) compete against each other, S_i wins if they can sing in every pitch that S_j can sing in, and some more pitches. Similarly, S_j wins if they can sing in every pitch that S_i can sing in, and some more pitches. If neither of these conditions are met, the match ends in a draw. **In this problem, you can assume that no match ends in a draw.**

N singers are competing in the tournament. Each singer competes in $N-1$ matches, one match against each of the other singers. The winner of a match scores 2 points, and the loser gets no points. But in case of a draw, both the singers get 1 point each.

You are given the lower and upper bounds of all the N singers. You need to output the total scores of each of the N singers at the end of the tournament.

Solution hint

Since no match ends in a draw, for any pair of singers S_i and S_j , one of their vocal ranges is strictly included in the other. Deduce that, across all singers, the vocal ranges form a sequence where each interval is strictly included in the previous one. You can then sort the starting points of the vocal ranges and determine how many matches each singer wins from the position of their starting point in this sorted sequence.

Input format

The first line of the input contains a single integer, N , which is the number of singers. N lines follow, the i -th of which contains two integers: L_i and U_i , which correspond to the lower bound and upper bound of the i -th singer.

Output format

Output a single line containing N integers, the i -th of which should be score of the i -th singer at the end of the tournament.

Test data

- $2 \leq N \leq 10^5$.
- $1 \leq L_i < U_i \leq 10^9$.
- All the $2N$ integers (lower bounds and upper bounds) are distinct.

- No matches end in a draw.

Sample input 1

```
5
3 23
4 20
11 16
5 19
1 25
```

Sample output 1

```
6 4 0 2 8
```

Sample input 2

```
7
3 22
9 17
6 19
13 16
2 25
14 15
5 21
```

Sample output 2

```
10 4 6 2 12 0 8
```

The due date for submitting this assignment has passed.

10 out of 10 tests passed.

You scored 100.0/100.

Assignment submitted on 2025-02-05, 22:37 IST

Your last recorded submission was :

```
1 #include <iostream>
2 #include <algorithm>
3 using namespace std;
4 int binarySearch(int *A,int n,int val)
5 {
6     int start=0;
7     int ends=n-1;
8     while(start<ends)
9     {
10         int mid=(start+ends)/2;
11         if(A[mid]==val)
12         {
13             return mid;
14         }
15         else if(A[mid]>val)
16         {
17             ends=mid-1;
18         }
19         else if(A[mid]<val)
20         {
21             start=mid+1;
22         }
23     }
24 }
25 int linearSearch(int *A,int n,int val)
26 {
27     for(int i=0;i<n;i++)
28     {
29         if(A[i]==val)
30         {
31             return i;
32         }
33     }
34 }
35 int main()
36 {
37     int N;
38     cin>>N;
39     int L[N],U[N],temp[N],dummy[N];
40     for(int i=0;i<N;i++)
41     {
42         cin>>L[i];
43         temp[i]=L[i];
44         cin>>U[i];
45     }
46     sort(L,L+N);
47     for(int i=0;i<N;i++)
48     {
49         int position=linearSearch(temp,N,L[i]);
50         dummy[position]=-(N-1-i)*2;
51     }
52     for(int i=0;i<N;i++)
53         cout<<dummy[i]<<" ";
54     cout<<endl;
55     return 0;
56 }
```


Week 3 Quiz

The due date for submitting this assignment has passed.

Due on 2025-02-12, 23:59 IST.

Score: 6/10=60%

Assignment submitted on 2025-02-11, 22:24 IST

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

- 1) An undirected graph G on 37 vertices has 5 connected components. What is the minimum number of edges in G ?

2 points

- ☐ 36
☒ 32
☐ 31
☐ Depends on the sizes of the five connected components.

Yes, the answer is correct.

Score: 2

Feedback:

A minimal connected graph on n vertices is a tree, with $n-1$ edges. If each of the five components is a tree and we join them with edges to form a single component, we would have to add 4 edges. The resulting graph would be a tree with 36 edges on 37 vertices. Hence, the four components originally had 32 edges. This is the same, regardless of how the graph is decomposed

Accepted Answers:

32

- 2) Suppose we have a directed graph $G = (V, E)$ with $V = \{1, 2, \dots, n\}$ and E presented as an adjacency list. For each vertex u in V , $L(u)$ is a list $[v_1, v_2, \dots, v_k]$ such that $(u, v_i) \in E$ for each i in $\{1, 2, \dots, k\}$.

2 points

If we reverse the edges in G , we get a new graph $G_R = (V, E_R)$ with the same set of vertices such that $(u, v) \in E_R$ if and only if $(v, u) \in E$.

We can represent G_R using an adjacency list where, for each u in V , $L_R(u)$ is the list of neighbours of u with respect to E_R .

Let n be the number of vertices in V and m be the number of edges in E . How long would it take to construct the adjacency lists $L_R(u)$, u in V , from the lists $L(u)$, u in V ?

- ☐ $O(m)$
☒ $O(n + m)$
☐ $O(n^2)$
☐ $O(n^2 + m)$

Yes, the answer is correct.

Score: 2

Feedback:

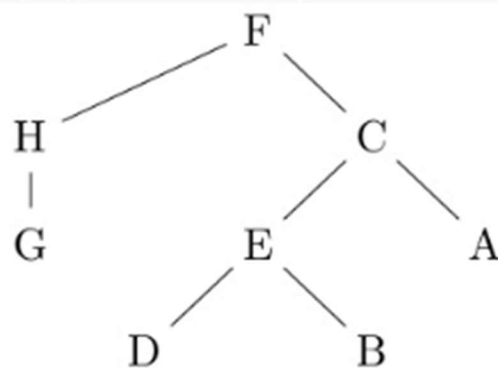
We can do it in $O(n+m)$ as follows. Initialize $L_R(u)$ to empty for each u in V (time $O(n)$). For each v in V , scan $L(v)$ and for each j in $L(v)$, add v to $L_R(j)$ (time $O(m)$ across all V). Note that $O(m)$ is not correct because if the edge set is empty, it still takes $O(n)$ time to initialize all the vertices.

Accepted Answers:

$O(n + m)$

- 3) Suppose we obtain the following DFS tree rooted at node F for an undirected graph G with vertices $\{A, B, C, D, E, F, G, H\}$.

2 points



Which of the following **cannot** be an edge in the graph G ?

- ☒ (F, G)
☐ (B, C)
☐ (A, F)
☐ (A, H)

No, the answer is incorrect.

Score: 0

Feedback:

In an undirected graph, all non-tree edges of a DFS tree must be along a path from an ancestor to a descendant. (A, H) is an edge across different paths and hence not possible.

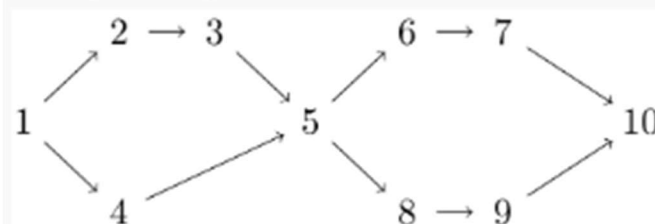
Accepted Answers:

(A, H)

4) We are interested in topological orderings of the following DAG that satisfy one or both of the following constraints:

- 4 appears before 3
- 8 appears after 7

How many such orderings are there?



- ☐ 9
☐ 13
☒ 16
☐ 18

No, the answer is incorrect.

Score: 0

Feedback:

Any topological sort is of the form 1, followed by a topological ordering of {2,3,4}, followed by 5, followed by a topological ordering {6,7,8,9}.

Nodes {2,3,4} have $3! = 6$ topological orderings of which only one, 2-3-4 has 3 before 4. Both other orderings have 4 before 3: 2-4-3, 4-2-3.

Nodes {6,7,8,9} have $4! = 24$ topological orderings, of which only 1 has 8 after 7, 6-7-8-9. All 5 others have 7 after 8: 6-8-7-9, 6-8-9-7, 8-6-7-9, 8-6-9-7, 8-9-6-7.

There are $3 \times 6 = 18$ total topological orderings of which $1 \times 5 = 5$ violates both conditions, so 13 satisfy one condition or both.

Accepted Answers:

13

5) Assembling a laptop consists of several steps, such as fixing the motherboard, inserting the battery, putting in the keyboard, attaching the screen, etc. Suppose there are 10 steps, labelled A, B, C, D, E, F, G, H, I, J. Each step takes a day to complete and we have the following dependencies between steps.

- A must happen before H
- B must happen before F
- B must happen before G
- C must happen before H
- D must happen before E
- E must happen before B
- F must happen before A
- F must happen before C
- G must happen before F
- I must happen before D
- I must happen before G
- J must happen before D
- J must happen before I

What is the minimum number of days required to complete the interiors?

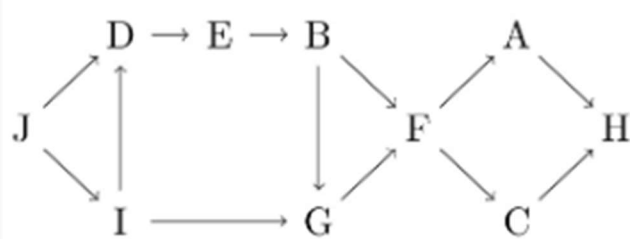
- ☒ 9
☐ 8
☐ 7
☐ 6

Yes, the answer is correct.

Score: 2

Feedback:

Here is the corresponding dag, whose longest path, J-I-D-E-B-G-F-A-H, for instance, has 9 vertices.



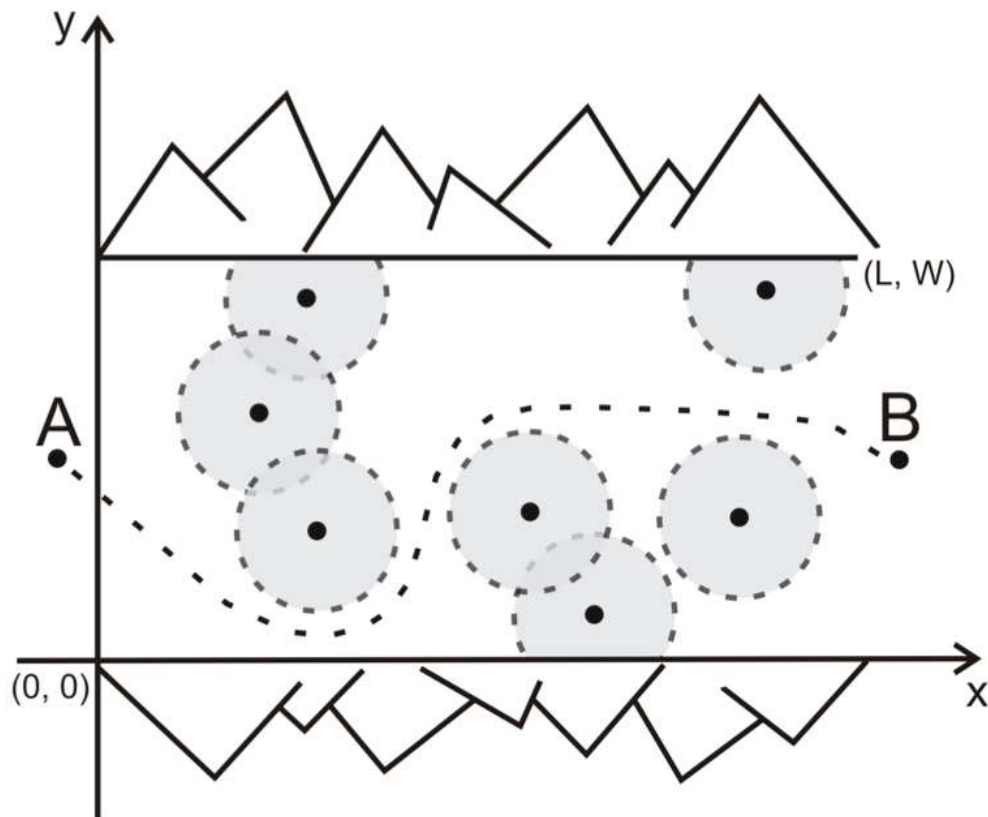
Accepted Answers:

9

Week 3 Programming Assignment - Prisoner Escape

Baltic Olympiad in Informatics, 2009)

A group of war prisoners are trying to escape from a prison. They have thoroughly planned the escape from the prison itself, and after that they hope to find shelter in a nearby village. However, the village (marked as B, see picture below) and the prison (marked as A) are separated by a canyon which is also guarded by soldiers. These soldiers sit in their pickets and rarely walk; the range of view of each soldier is limited to exactly 100 meters. Thus, depending on the locations of soldiers, it may be possible to pass the canyon safely, keeping the distance to the closest soldier strictly larger than 100 meters at any moment.



You are to write a program which, given the width and the length of the canyon and the coordinates of every soldier in the canyon, and assuming that soldiers do not change their locations, determines whether prisoners can pass the canyon unnoticed.

Solution Hint

The soldiers can be modelled as an undirected graph G . Let each soldier be represented by a vertex. Add an edge between two vertices if and only if the range of view of the corresponding soldiers overlaps. Add two additional vertices s and t , representing the northern and southern side of the canyon, respectively. Connect s and t to those vertices representing soldiers whose range of view includes the respective side of the canyon. Use depth-first-search or breadth-first-search to check whether there is a path between s and t in G .

Input format

The first line contains three integers L , W , and N – the length and the width of the canyon, and the number of soldiers, respectively. Each of the following N lines contains a pair of integers X_i and Y_i – the coordinates of i -th soldier in the canyon ($0 \leq X_i \leq L$, $0 \leq Y_i \leq W$). The coordinates are given in meters, relative to the canyon: the southwestern corner of the canyon has coordinates $(0, 0)$, and the northeastern corner of the canyon has coordinates (L, W) , as seen in the picture above. Note that passing the canyon may start at coordinate $(0, y_s)$ for any $0 \leq y_s \leq W$ and end at coordinate (L, y_e) for any $0 \leq y_e \leq W$. Neither y_s nor y_e need to be integer.

Output format

Output a single integer: 0 if the prisoners can escape, 1 if they cannot.

Test data

$1 \leq W \leq 50,000$ $1 \leq L \leq 50,000$ $1 \leq N \leq 250$

Example

Sample input 1

```
130 340 5
10 50
130 130
70 170
```

0 180
60 260

Sample output 1

1

Sample input 2

500 300 4
250 0
250 300
100 150
400 150

Sample output 2

0

The due date for submitting this assignment has passed.
20 out of 20 tests passed.
You scored 100.0/100.

Assignment submitted on 2025-02-13, 10:19 IST

Your last recorded submission was :

```
1 import sys
2 from collections import deque
3
4 def main():
5     input = sys.stdin.read().split()
6     idx = 0
7     L = int(input[idx]); idx += 1
8     W = int(input[idx]); idx += 1
9     N = int(input[idx]); idx += 1
10
11     soldiers = []
12     for _ in range(N):
13         x = int(input[idx]); idx += 1
14         y = int(input[idx]); idx += 1
15         soldiers.append((x, y))
16
17     adj = [[] for _ in range(N + 2)]
18     s = N
19     t = N + 1
20
21     for i in range(N):
22         x, y = soldiers[i]
23         if y <= 100:
24             adj[i].append(s)
25             adj[s].append(i)
26         if (W - y) <= 100:
27             adj[i].append(t)
28             adj[t].append(i)
29
30     for i in range(N):
31         x1, y1 = soldiers[i]
32         for j in range(i + 1, N):
33             x2, y2 = soldiers[j]
34             dx = x1 - x2
35             dy = y1 - y2
36             dist_sq = dx * dx + dy * dy
37             if dist_sq <= 200 * 200:
38                 adj[i].append(j)
39                 adj[j].append(i)
40
41     visited = [False] * (N + 2)
42     q = deque()
43     q.append(s)
44     visited[s] = True
45     found = False
46
47     while q:
48         u = q.popleft()
49         if u == t:
50             found = True
51             break
52         for v in adj[u]:
53             if not visited[v]:
54                 visited[v] = True
55                 q.append(v)
56
57     print(1 if found else 0)
58
59 if __name__ == "__main__":
60     main()
```

Week 4 Quiz

The due date for submitting this assignment has passed.

Due on 2025-02-19, 23:59 IST.

Score: 6/10=60%

Assignment submitted on 2025-02-19, 22:32 IST

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

1) A regional airline serves 10 cities. It operates to-and-fro return flights between 9 pairs of cities in such a way that every city is reachable from every other city through a sequence of connecting flights. **2 points**

We know the fuel consumption for the direct flights between each pair of cities. We want to compute the minimum fuel consumption from a given city to every other city in the airline network. Which of the following is true for this specific situation?

- ☐ We can use BFS, DFS or Dijkstra's algorithm to compute this.
- ☐ We can use BFS or Dijkstra's algorithm to compute this, but not DFS.
- ☐ We can use DFS or Dijkstra's algorithm to compute this, but not BFS.
- ☒ We can only use Dijkstra's algorithm to compute this, not BFS or DFS.

No, the answer is incorrect.

Score: 0

Feedback:

The network is a weighted tree, so there is only one path between any pair of nodes. Hence BFS and DFS will work as well as Dijkstra's algorithm from any starting point.

Accepted Answers:

We can use BFS, DFS or Dijkstra's algorithm to compute this.

2) An airline charges a fixed price for each direct flight. For each sequence of hopping flights, the ticket price is the sum of the fares of the individual sectors. **2 points**

TripGuru has precalculated the cheapest routes between all pairs of cities on this airline's route network so that it can offer an optimum choice instantly to customers visiting its website.

The government decides to impose a 3% luxury tax on each sector. Which of the following describes the impact of this surcharge on TripGuru's computation?

- ☒ There is no impact. Cheapest routes between all pairs of cities remains unchanged.
- ☐ The surcharge favours hopping flights with more sectors. TripGuru should recompute any cheapest route where there is a longer route in terms of number of flights.
- ☐ The surcharge favours hopping flights with fewer sectors. TripGuru should recompute any cheapest route where there is a shorter route in terms of number of flights.
- ☐ The impact is unpredictable. TripGuru should recompute all cheapest routes.

Yes, the answer is correct.

Score: 2

Feedback:

For any path, the total cost increases by 3%, regardless of the length of the path, so the current cheapest routes remain unchanged.

Accepted Answers:

There is no impact. Cheapest routes between all pairs of cities remains unchanged.

3) An airline charges a fixed price for each direct flight. For each sequence of hopping flights, the ticket price is the sum of the fares of the individual sectors. **2 points**

TripGuru has precalculated the cheapest routes between all pairs of cities on this airline's route network. A major IT company has its offices at all the locations served by the airline, and books all official trips for its employees via TripGuru. To simplify administrative processing, the IT company has selected a subset of the airline's routes so that all of their office locations are reachable from each other, and the total cost of travelling across all the chosen routes is minimum.

The airline has decided to become a full-service carrier and has included a meal on each sector. To account for this, the airline has added a flat "convenience fee" of Rs 300 on each sector. Which of the following describes the impact of this surcharge on the IT company's computation of which subset of routes to use?

- ☐ There is no impact. The IT company can retain the same subset of routes.
- ☐ The surcharge favours hopping flights with more sectors. The IT company should reconsider including routes where there may be a longer route, with more hops.
- ☐ The surcharge favours hopping flights with fewer sectors. The IT company should reconsider including routes where there may be a shorter route, with fewer hops.
- ☒ The effect of the surcharge is unpredictable. The IT company should recompute its preferred subset of routes afresh.

No, the answer is incorrect.

Score: 0

Feedback:

The IT company computes a minimum cost spanning tree. Any spanning tree over n nodes has $n-1$ edges. So the cost of any spanning increases by the same amount, $300(n-1)$, after the surcharge. Hence, the MCST remains unchanged.

Accepted Answers:

There is no impact. The IT company can retain the same subset of routes.

4) Suppose we have a weighted undirected graph with negative edge weights. Which of the following is correct? **2 points**

- ☐ Neither Kruskal's algorithm nor Prim's algorithm can be used to compute an MCST.
- ☐ Kruskal's algorithm will compute a valid MCST but Prim's algorithm will not.
- ☐ Prim's algorithm will compute a valid MCST but Kruskal's algorithm will not.
- ☒ Both Kruskal's algorithm and Prim's algorithm can be used to compute an MCST.

Yes, the answer is correct.

Score: 2

Feedback:

Both Kruskal's algorithm and Prim's algorithm are unaffected by negative edge weights.

Accepted Answers:

Both Kruskal's algorithm and Prim's algorithm can be used to compute an MCST.

5) How can we use the Floyd-Warshall algorithm for all-pairs shortest paths to detect whether a graph has a negative cycle? 2 points

- ☐ Check if any shortest path entry $A[i][i]$ is negative.
- ☒ Check if any shortest path entry $A[i][i]$ on the diagonal is negative.
- ☐ Check if any shortest path entry $A[i][i]$ reduces from one iteration to the next.
- ☐ The Floyd-Warshall algorithm cannot be used to detect negative cycles.

Yes, the answer is correct.
 Score: 2
 Feedback:
 Without a negative cycle, $A[i][i]$ should remain 0 till the end for all i . If there is a negative cycle involving i , $A[i][i]$ will become negative.
 Accepted Answers:
 Check if any shortest path entry $A[i][i]$ on the diagonal is negative.

Week 4 Programming Assignment - Number Triples

(IARCS OPC Archive, K Narayan Kumar, CMI)

In this problem you will be given a sequence of triples of positive integers. For example:

1	2	5
5	2	6
1	3	8
8	1	11
1	1	6
10	1	6
11	3	6
10	4	8
10	1	11

Given a pair of numbers A and B, a chain connecting A and B is a sequence of triples

$$A_0 \ W_0 \ A_1, \ A_1 \ W_1 \ A_2, \ A_2 \ W_2 \ A_3, \ \dots \ A_{k-2} \ W_{k-2} \ A_{k-1}, \ A_{k-1} \ W_{k-1} \ A_k$$

such that

$$A_0 = A$$

$$A_k = B$$

For each i , $0 \leq i < k$, either the triple $A_i \ W_i \ A_{i+1}$ or the triple $A_{i+1} \ W_i \ A_i$ is in the given set of triples.

The value of such a chain is the sum of the W_i s in the chain. For example, here is a chain connecting 1 and 11 using the triples listed above:

1	1	6,	6	3	11
---	---	----	---	---	----

The value of this chain is $1+3 = 4$.

Here is another chain connecting 1 and 11.

1	1	6,	6	1	10,	10	1	11
---	---	----	---	---	-----	----	---	----

The value of this chain is $1+1+1 = 3$. You can verify that among all chains connecting 1 and 11 this is the one with least value.

Sometimes there may be no chains connecting the given pair of numbers. For example, there is no chain connecting 1 and 2.

You will be given a sequence of triples and a pair of numbers. Your task is to find the value of the least value chain connecting the two numbers.

Solution hint

Construct a graph from the triples and use Dijkstra's algorithm for single-source shortest paths.

Input format

The first line of the input has three numbers M, A and B. M is the number of triples. The next M lines (lines 2,3,...,M+1) describe the triples. Line $i+1$ contains the three positive integers X_i , Y_i and Z_i that make up the i th triple. Your task is to find the value of the least value chain connecting A and B.

Output format

If there is at least one chain connecting A and B the first line of the output must consist of a single word `YES`. In that case the second line must contain a single integer value indicating the value of the least valued chain connecting A and B. If there are no chains connecting A and B the output should contain a single line with the word `NO` on it.

Test Data:

You may assume that $1 \leq X_i, Y_i, Z_i \leq 2000$ and $M \leq 4000000$.

Example:

We illustrate the input and output format using the above example:

Sample Input 1:

```
9 1 11
1 2 5
5 2 6
1 3 8
8 1 11
1 1 6
10 1 6
11 3 6
10 4 8
10 1 11
```

Sample Output 1:

```
YES
3
```

Sample Input 2:

```
9 1 2
1 2 5
5 2 6
1 3 8
8 1 11
1 1 6
10 1 6
11 3 6
10 4 8
10 1 11
```

Sample Output 2:

```
NO
```


Assignment submitted on 2025-02-19, 22:32 IST

Your last recorded submission was :

```
1 from collections import defaultdict
2 import heapq
3
4 def build_graph(triples):
5     graph = defaultdict(list)
6
7     for x, y, z in triples:
8         graph[x].append((z, y))
9         graph[z].append((x, y))
10
11     return graph
12
13 def dijkstra(graph, start, end):
14     pq = [(0, start)]
15     distances = defaultdict(lambda: float('inf'))
16     distances[start] = 0
17     visited = set()
18
19     while pq:
20         curr_dist, curr_node = heapq.heappop(pq)
21
22         if curr_node in visited:
23             continue
24
25         visited.add(curr_node)
26
27         if curr_node == end:
28             return curr_dist
29
30         for next_node, weight in graph[curr_node]:
31             if next_node in visited:
32                 continue
33
34             distance = curr_dist + weight
35
36             if distance < distances[next_node]:
37                 distances[next_node] = distance
38                 heapq.heappush(pq, (distance, next_node))
39
40     return None
41
42 def solve_number_triples(M, A, B, triples):
43     graph = build_graph(triples)
44     result = dijkstra(graph, A, B)
45
46     if result is None:
47         return False, None
48     else:
49         return True, result
50
51 def main():
52     M, A, B = map(int, input().split())
53     triples = []
54     for _ in range(M):
55         x, y, z = map(int, input().split())
56         triples.append((x, y, z))
57
58     path_exists, min_value = solve_number_triples(M, A, B, triples)
59
60     if path_exists:
61         print("YES")
62         print(min_value)
```

Week 5 Quiz

The due date for submitting this assignment has passed.

Due on 2025-02-26, 23:59 IST.

Score: 10/10=100%

Assignment submitted on 2025-02-26, 19:51 IST

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

1) An image is represented as an $N \times N$ array A of pixels. There is a function that transforms the image as follows:

2 points

- The function scans each pixel $A[i][j]$. Depending on the current value of $A[i][j]$, some values in row i and column j are updated. It is possible that all values in row i and column j are updated. It is also possible that no values are updated.
- Updating a pixel takes time $O(\log N)$.
- Each pixel in the image is updated at most once by the function.

What is the best upper bound you can estimate for the total time taken across all the updates made by the function?

- ☐ $O(N^2)$
- ☒ $O(N^2 \log N)$
- ☐ $O(N^3)$
- ☐ $O(N^3 \log N)$

Yes, the answer is correct.

Score: 2

Feedback:

A naive analysis says that we update upto $O(N)$ entries when scanning each $A[i][j]$. Thus the update at each $A[i][j]$ takes $O(N \log N)$ in the worst case, so $O(N^2 \log N)$ overall, since there are N^2 entries in the array.

However, since each of the N^2 entries is updated at most once, an amortized analysis says that the total time cannot exceed $O(N^2 \log N)$.

Accepted Answers:

$O(N^2 \log N)$

2) In the Union-Find data structure, suppose we want to add an operation `reassign(j, k)` that moves item `j` to the same component as item `k`. **2 points**
What would be the complexity of this operation for a set with `n` elements?

- ☐ $O(1)$ for both the array representation and the pointer representation.
- ☐ $O(1)$ for the array representation, but not $O(1)$ for the pointer representation.
- ☐ $O(1)$ for the pointer representation, but not $O(1)$ for the array representation.
- ☒ Not $O(1)$ for either the array representation or the pointer representation.

Yes, the answer is correct.

Score: 2

Feedback:

In the array representation, we have to assign `Component[j]` the same value as `Component[k]`. In the pointer representation, we look up `Node[j]` and `Node[k]` and copy the parent pointer from `Node[k]` to `Node[j]`. So both these updates take time $O(1)$.

However, in the array representation we also have to update `members[i]` for both components, which will take time proportional to the sizes of the components, so it is not a constant time operation.

In the pointer representation we also have to shift the pointer of the children of `j` to the parent of `j`. Since `j` could have multiple children, this is again not a constant time operation.

Accepted Answers:

Not $O(1)$ for either the array representation or the pointer representation.

3) In a min-heap, what is the most accurate description of the worst-case complexity of the operation `find_max` that reports the value of the largest element in the heap, without removing it? **2 points**

- ☐ $O(1)$
- ☐ $O(\log n)$
- ☒ $O(n)$
- ☐ $O(n \log n)$

Yes, the answer is correct.

Score: 2

Feedback:

The min-heap structure does not help us find the maximum value in any way, so the simplest option is to scan the heap as an unsorted array and return the maximum value in $O(n)$ time.

Accepted Answers:

$O(n)$

4) After inserting 57 into a max-heap, the structure of the heap is [82, 57, 27, 42, 25, 18, 25, 27, 32]. What was the structure of the heap before inserting 57? **2 points**

- ☐ [82, 42, 32, 27, 25, 18, 25, 27]
- ☐ [82, 27, 42, 32, 25, 18, 25, 27]
- ☐ [82, 42, 27, 25, 32, 18, 25, 27]
- ☒ [82, 42, 27, 32, 25, 18, 25, 27]

Yes, the answer is correct.

Score: 2

Feedback:

The leaf node where 57 was inserted was the last one, which is currently 32. The path from the current position of 57 to this leaf goes via the node with value 42. So we have to undo the swap made along this path when inserting 57.

Accepted Answers:

[82, 42, 27, 32, 25, 18, 25, 27]

5) Consider an alternative to binary search called ternary search that divides the input array into three equal parts and recursively searches one of these three segments. What can we say about the asymptotic worst case complexity of ternary search versus binary search? **2 points**

- ☒ The complexity of ternary search is the same as that of binary search.
- ☐ The complexity of binary search is strictly better than that of ternary search.
- ☐ The complexity of ternary search is strictly better than that of binary search.
- ☐ The relative complexity of ternary and binary search depends on the distribution of values across the array.

Yes, the answer is correct.

Score: 2

Feedback:

The recurrence for ternary search is

- $T(0) = T(1) = 1$
- $T(n) = O(1) + T(n/3)$

which resolves as $T(n) = O(\log_3 n)$. However, since $\log_3 n = \log_2 n / \log_2 3$, the asymptotic complexity is the same as $O(\log_2 n)$ for binary search.

Accepted Answers:

The complexity of ternary search is the same as that of binary search.

Week 6 Quiz

The due date for submitting this assignment has passed.

Due on 2025-03-05, 23:59 IST.

Score: 6/8=75%

Assignment submitted on 2025-02-26, 13:17 IST

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

1) Suppose we implement a binary search tree without a parent pointer in each node. So each node has a value and pointers to the left and right children. What would be the complexity of computing the successor for a node in such an implementation for a balanced search tree on n nodes? **2 points**

- ☐ $O(n)$ for all nodes
- ☐ $O(\log n)$ for all nodes
- ☒ $O(\log n)$ for a node with a right child, $O(n)$ otherwise
- ☐ $O(\log n)$ for a node with a right child, $O(n \log n)$ otherwise

No, the answer is incorrect.

Score: 0

Feedback:

To find the successor without parent pointers, we start a second scan from the root and mark the point where we turn left to enter the subtree containing the current node, so it takes $O(\log n)$ in all cases.

Accepted Answers:

$O(\log n)$ for all nodes

2) We have n distinct values stored in a height balanced (AVL) binary search tree. Which of the following statements is *always* true? **2 points**

- ☐ The value at each node is the median of the values in the subtree rooted at that node.
- ☒ The shortest path between any pair of nodes is at most $O(\log n)$.
- ☐ For any node, the difference between the size of the left subtree and the right subtree is at most 3.
- ☐ The number of leaf nodes is greater than or equal to the number of internal nodes.

Yes, the answer is correct.

Score: 2

Feedback:

A height balanced tree has overall height at most $O(\log n)$, so the shortest path between any pair of nodes is always at most $O(\log n)$.

The following AVL tree is a counterexample for all other statements.

3) Postorder traversal lists out the nodes in a binary tree as follows. **2 points**

- First do a postorder traversal of the left subtree of the root.
- Then do a postorder traversal of the right subtree of the root.
- Finally list out the value at the root.

The postorder traversal of a binary search tree with integer values produces the following sequence: 17, 28, 25, 51, 98, 55, 42, 37. What is the value of the left child of the root of the tree?

- ☐ 37
- ☐ 28
- ☒ 25
- ☐ 17

Yes, the answer is correct.

Score: 2

4) Consider the following function to traverse a search tree t with integer values, where $\text{prime}(m)$ returns True if m is an even number and False otherwise. **2 points**

```
function strangeOrder(t) {
  if (t != NIL) {
    if (prime(t.value)){
      strangeOrder(t.left);
      print(t.value);
      strangeOrder(t.right);
    }else{
      strangeOrder(t.right);
      print(t.value);
      strangeOrder(t.left);
    }
  }
}
```

What is the complexity of this traversal for a binary search tree with n nodes?

- ☒ $O(n)$ whether the tree is balanced or unbalanced.
- ☐ $O(n \log n)$ whether the tree is balanced or unbalanced.
- ☐ $O(\log n)$ if the tree is balanced, $O(n)$ otherwise.
- ☐ $O(n)$ if the tree is balanced, $O(n \log n)$ otherwise.

Yes, the answer is correct.

Score: 2

5) Suppose we build a Huffman code over the four letter alphabet {a,b,c,d}, where $f(a)$, $f(b)$, $f(c)$ and $f(d)$ denote the frequencies (probabilities) of the letters and $f(a) > f(b) > f(c) > f(d)$. Which of the following conditions will result in a Huffman code assigning a 2 bit code for each letter? 0 points

- ☐ $f(a) > f(c)+f(d)$
- ☐ $f(b) > f(c)+f(d)$
- ☒ $f(a) > f(b)+f(c)+f(d)$
- ☐ It is not possible to have a Huffman code that assigns two bits to each letter.

No, the answer is incorrect.

Score: 0

Feedback:

There is an error in this question and none of the options are correct. Hence, this question is being assigned zero weightage and the assignment will be reevaluated to only consider the other four questions.

The correct answer is $f(a) < f(c)+f(d)$.

The first iteration will group {c,d} as {cd}. To assign 2 bits per letter, the second iteration should combine {a,b} to form {ab}, rather than combining {b} with {cd} to form {bcd}. For this, we need $f(a)+f(b) < f(b)+f(c)+f(d)$, which simplifies to $f(a) < f(c)+f(d)$.

Accepted Answers:

It is not possible to have a Huffman code that assigns two bits to each letter.

Week 6 Programming Assignment - Siruseri Sport Stadium

(IARCS OPC Archive, K Narayan Kumar, CMI)

The bustling town of Siruseri has just one sports stadium. There are a number of schools, colleges, sports associations, etc. that use this stadium as the venue for their sports events.

Anyone interested in using the stadium has to apply to the Manager of the stadium indicating both the starting date (a positive integer S) and the length of the sporting event in days (a positive integer D) they plan to organise. Since these requests could overlap it may not be possible to satisfy everyone.

It is the job of the Manager to decide who gets to use the stadium and who does not. The Manager, being a genial man, would like to keep as many organisations happy as possible and hence would like to allocate the stadium so that maximum number of events are held.

Suppose, for example, the Manager receives the following 4 requests:

Event No.	Starting Date	Length
1	2	5
2	9	7
3	15	6
4	9	3

He would allot the stadium to events 1, 4 and 3. Event 1 begins on day 2 and ends on day 6, event 4 begins on day 9 and ends on day 11 and event 3 begins on day 15 and ends on day 20. You can verify that it is not possible to schedule all the 4 events (since events 2 and 3 overlap and only one of them can get to use the stadium).

Your task is to help the manager find the best possible allotment (i.e., the maximum number of events that can use the stadium).

Solution hint

Use the greedy algorithm for interval scheduling.

Input format

The first line of the input will contain a single integer N ($N \leq 100000$) indicating the number of events for which the Manager has received a request. Lines 2,3,...,N+1 describe the requirements of the N events. Line $i+1$ contains two integers S_i and D_i indicating the starting date and the duration of event i . You may assume that $1 \leq S_i \leq 1000000$ and $1 \leq D_i \leq 1000$.

Output format

Your output must consist of a single line containing a single integer M , indicating the maximum possible number of events that can use the stadium.

Constraints

The range of values over which your program is to be tested is mentioned above. In addition, 50% of the test cases will also satisfy $N \leq 10000$.

Example:

We now illustrate the input and output formats using the example described above.

Sample input:

```
4
2 5
9 7
15 6
9 3
```

Sample output:

```
3
```

The due date for submitting this assignment has passed.

10 out of 10 tests passed.

You scored 100.0/100.

Assignment submitted on 2025-02-26, 13:30 IST

Your last recorded submission was :

```
1 #include <iostream>
2 #include <vector>
3 #include <algorithm>
4
5 using namespace std;
6
7 struct Event {
8     int start;
9     int duration;
10    int end;
11 };
12
13 bool compareByEndDate(const Event &a, const Event &b) {
14     return a.end < b.end;
15 }
16
17 int main() {
18     int n;
19     cin >> n;
20
21     vector<Event> events(n);
22
23     for (int i = 0; i < n; i++) {
24         cin >> events[i].start >> events[i].duration;
25         events[i].end = events[i].start + events[i].duration - 1;
26     }
27
28     sort(events.begin(), events.end(), compareByEndDate);
29
30     int count = 0;
31     int lastEndDate = 0;
32
33     for (int i = 0; i < n; i++) {
34         if (events[i].start > lastEndDate) {
35             count++;
36             lastEndDate = events[i].end;
37         }
38     }
39
40     cout << count << endl;
41
42     return 0;
43 }
```


Week 7 Quiz

The due date for submitting this assignment has passed.

Due on 2025-03-12, 23:59 IST.

Score: 10/10=100%

Assignment submitted on 2025-03-05, 20:25 IST

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

Your final exams are over and you are catching up on sports on TV. You have a schedule of interesting matches from all over the world during the next week. You hate to start or stop watching a match midway, so your aim is to watch as many complete matches as possible during the week.

Suppose there are n such matches $\{M_1, M_2, \dots, M_n\}$ available during the coming week. The matches are ordered by starting time, so for each $i \in \{1, 2, \dots, n-1\}$, M_i starts before M_{i+1} . However, match M_i may not end before M_{i+1} starts, so for each $i \in \{1, 2, \dots, n-1\}$, $\text{Next}[i]$ is the smallest $j > i$ such that M_j starts after M_i finishes.

Given the sequence $\{M_1, M_2, \dots, M_n\}$ and the values $\text{Next}[i]$ for each $i \in \{1, 2, \dots, n-1\}$, your aim is to compute the maximum number of complete matches that can be watched.

1) Let $\text{Watch}[i]$ denote the maximum number of complete matches that can be watched among $\{M_i, M_{i+1}, \dots, M_n\}$. Which of the following is a correct recursive formulation of $\text{Watch}[i]$?

- ☐ $\text{Watch}[1] = 1$
 $\text{Watch}[i] = \max(\text{Watch}[i-1] + 1, \text{Watch}[\text{Next}[i-1]]), i \in \{2, 3, \dots, n\}$
- ☒ $\text{Watch}[n] = 1$
 $\text{Watch}[i] = \max(1 + \text{Watch}[\text{Next}[i]], \text{Watch}[i+1]), i \in \{1, 2, \dots, n-1\}$
- ☐ $\text{Watch}[1] = 1$
 $\text{Watch}[i] = \max(\text{Watch}[i-1], 1 + \text{Watch}[\text{Next}[i-1]]), i \in \{2, 3, \dots, n\}$
- ☐ $\text{Watch}[n] = 1$
 $\text{Watch}[i] = \max(\text{Watch}[\text{Next}[i]], 1 + \text{Watch}[i+1]), i \in \{1, 2, \dots, n-1\}$

Yes, the answer is correct.

Score: 2

2) What is the size of the memo table for this problem?

2 points

- ☐ $N+1$
- ☒ N
- ☐ $N-1$
- ☐ N^2

Yes, the answer is correct.

Score: 2

Feedback:

One entry per day, N days.

Accepted Answers:

N

3) What is a good order to compute $\text{Watch}[i]$ using dynamic programming?

2 points

- ☒ From $\text{Watch}[n]$ to $\text{Watch}[1]$
- ☐ From $\text{Watch}[1]$ to $\text{Watch}[n]$
- ☐ Either from $\text{Watch}[1]$ to $\text{Watch}[n]$ or from $\text{Watch}[n]$ to $\text{Watch}[1]$
- ☐ None of these

Yes, the answer is correct.

Score: 2

Feedback:

The base case is $\text{Watch}[n]$, so start with $\text{Watch}[n]$ and work backwards to $\text{Watch}[1]$.

Accepted Answers:

From $\text{Watch}[n]$ to $\text{Watch}[1]$

4) How much time will it take to compute $\text{Watch}[1]$ using dynamic programming?

2 points

- ☐ $O(n^3)$
- ☐ $O(n^2)$
- ☐ $O(n \log n)$
- ☒ $O(n)$

Yes, the answer is correct.

Score: 2

5) Suppose the list of matches to be watched is presented in the form

$[(7,45), (15,31), (35,47), (46,61), (48,60), (57,58), (59,63), (64,70), (71,80), (75,90), (81,83), (91,100)]$

where each match M_i is represented by a pair (S_i, T_i) indicated its starting time and ending time.

To be able to watch both M_i and M_j , for $j > i$, it must be the case that $S_j > T_i$.

What is the maximum number of matches you can watch in this case?

8

Yes, the answer is correct.

Score: 2

Week 7 Programming Assignment - IOI Training Camp 20xx

We are well into the 21st century and school children are taught dynamic programming in class 4. The IOI training camp has degenerated into an endless sequence of tests, with negative marking. At the end of the camp, each student is evaluated based on the sum of the best *contiguous segment* (i.e., *no gaps*) of marks in the overall sequence of tests.

Students, however, have not changed much over the years and they have asked for some relaxation in the evaluation procedure. As a concession, the camp coordinators have agreed that students are allowed to drop upto a certain number of tests when calculating their best segment.

For instance, suppose that Lavanya is a student at the training camp and there have been ten tests, in which her marks are as follows.

Test	1	2	3	4	5	6	7	8	9	10
Marks	6	-5	3	-7	6	-1	10	-8	-8	8

In this case, without being allowed to drop any tests, the best segment is tests 5–7, which yields a total of 15 marks. If Lavanya is allowed to drop upto 2 tests in a segment, the best segment is tests 1–7, which yields a total of 24 marks after dropping tests 2 and 4. If she is allowed to drop upto 6 tests in a segment, the best total is obtained by taking the entire list and dropping the 5 negative entries to get a total of 33.

You will be given a sequence of N test marks and a number K . You have to compute the sum of the best segment in the sequence when upto K marks may be dropped from the segment.

Solution hint

For $1 \leq i \leq N$, $1 \leq j \leq K$, let $\text{Best}[i][j]$ denote the maximum segment ending at position i with at most j marks dropped. $\text{Best}[i][0]$ is the classical maximum subsegment or maximum subarray problem. For $j \geq 1$; inductively compute $\text{Best}[i][j]$ from $\text{Best}[i][j-1]$.

Input format

The first line of input contains two integers N and K , where N is the number of tests for which marks will be provided and K is the limit of how many entries may be dropped from a segment.

This is followed by N lines of input each containing a single integer. The marks for test i , $i \in \{1, 2, \dots, N\}$ are provided in line $i+1$.

Output format

The output is a single number, the maximum marks that can be obtained from a segment in which upto K values are dropped.

Constraints

You may assume that $1 \leq N \leq 10^4$ and $0 \leq K \leq 10^2$. The marks for each test lie in the range $[-10^4 \dots 10^4]$. In 40% of the cases you may assume $N \leq 250$.

Example:

We now illustrate the input and output formats using the example described above.

Sample input:

```
10 2
6
-5
3
-7
6
```


-1
10
-8
-8
8

Sample output:

24

The due date for submitting this assignment has passed.

10 out of 10 tests passed.

You scored 100.0/100.

Assignment submitted on 2025-03-04, 17:25 IST

Your last recorded submission was :

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     int n, k;
6     cin >> n >> k;
7     vector<int> marks(n);
8     for (int i = 0; i < n; i++) {
9         cin >> marks[i];
10    }
11
12    vector<vector<int>> best(n + 1, vector<int>(k + 1, INT_MIN));
13    vector<vector<int>> curr(n + 1, vector<int>(k + 1, 0));
14
15    int result = INT_MIN;
16
17    for (int i = 0; i < n; i++) {
18        for (int j = 0; j <= k; j++) {
19            if (i == 0) {
20                best[i][j] = marks[i];
21                curr[i][j] = marks[i];
22                continue;
23            }
24
25            if (j == 0) {
26                curr[i][j] = max(curr[i-1][j] + marks[i], marks[i]);
27                best[i][j] = max(best[i-1][j], curr[i][j]);
28                continue;
29            }
30
31            curr[i][j] = max({
32                curr[i-1][j] + marks[i],
33                curr[i-1][j-1],
34                marks[i]
35            });
36
37            best[i][j] = max(best[i-1][j], curr[i][j]);
38        }
39    }
40
41    for (int j = 0; j <= k; j++) {
42        result = max(result, best[n-1][j]);
43    }
44
45    cout << result << endl;
46    return 0;
47 }
```

Week 8 Quiz

The due date for submitting this assignment has passed.

Due on 2025-03-19, 23:59 IST.

Score: 8/10=80%

Assignment submitted on 2025-03-16, 19:39 IST

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

1) Which of the following is a linear constraint?

2 points

- ☐ $17x + 3xz \leq 4$
- ☒ $3x \geq 14y + 2z + 13$
- ☐ $7x \leq 3xy + 14z - 12$
- ☐ $5y + 3x^2 \geq 33$

Yes, the answer is correct.

Score: 2

Feedback:

In a non-linear constraint, we have two or more variables (same or different) multiplied together.

Accepted Answers:

$3x \geq 14y + 2z + 13$

2) A critical document has to be sent from Ougadougou to Timbuktu. To ensure that it reaches safely, three copies are made and entrusted to three different people to carry to the destination. They will each take a sequence of trains, buses and flights. To ensure that the document reaches safely, the three couriers are not allowed to take the same train, bus or flight at any stage along the route.

This can be modelled as a network flow problem where the source and target are the Ougadougou and Timbuktu, the cities along the way are nodes and each direct connection between two cities by train, bus or plane is an edge. The actual flow problem to be solved is to:

- ☐ Assign a total of capacity 3 to all outgoing edges from the source and find a feasible flow.
- ☐ Assign a total of capacity 3 to all incoming edges to the target and find a feasible flow.
- ☐ Assign each edge capacity 1 and check that the maximum flow is less than 3.
- ☒ Assign each edge capacity 1 and check that the maximum flow is at least 3.

Yes, the answer is correct.

Score: 2

3) A major political party's main convention is currently in progress. The biggest 24 hour news channel in town wants full video coverage of all the major events. Unfortunately, some of these events overlap with each other. The channel's programme managers are trying to compute the minimum number of camera teams needed to cover all the relevant events.

The programme managers decide to model this as a graph where the nodes are the major events to be covered and edges represent pairs of events with overlapping timings. In this setting, the graph theoretic question to be answered is:

- ☐ Find a spanning tree with minimum number of edges.
- ☐ Find a minimum size decomposition into connected components.
- ☒ Find a minimal colouring.
- ☐ Find a minimum size vertex cover.

Yes, the answer is correct.

Score: 2

Feedback:

Colours represent camera teams. A valid colouring ensures that overlapping events are covered by different camera teams. A minimal colouring minimizes the number of camera teams needed.

Accepted Answers:

Find a minimal colouring.

4) We wish to show that problem B is NP-complete. Which of the following facts is sufficient to establish this.

2 points

- ☐ There is a polynomial time reduction from B to SAT.
- ☐ There is a polynomial time reduction from SAT to B
- ☐ There is a polynomial time reduction from B to SAT, and B has a checking algorithm.
- ☒ There is a polynomial time reduction from SAT to B, and B has a checking algorithm.

Yes, the answer is correct.

Score: 2

5) We have constructed a polynomial time reduction from problem A to problem B. Which of the following is a valid inference?

2 points

- ☐ If the best algorithm for A takes exponential time, there is no polynomial time algorithm for B.
- ☐ If we have a polynomial time algorithm for A, we must also have a polynomial time algorithm for B.
- ☐ If we don't know whether there is a polynomial time algorithm for B, there cannot be a polynomial time algorithm for A.
- ☒ If the best algorithm for B takes exponential time, there is no polynomial time algorithm for A.

No, the answer is incorrect.

Score: 0

Feedback:

If we have a polynomial time solution for B, the reduction gives us a polynomial time solution for A.

Accepted Answers:

If the best algorithm for A takes exponential time, there is no polynomial time algorithm for B.

Unit wise Progress

NOTE: The latest Course and Unit progress will be reflected within 24 hours.

Unit	Lesson Progress	Assignments Progress
About NPTEL	0.0%	-
How does an NPTEL online course work?	100.0%	-
Week 1 : Introduction	100.0%	-
Week 1 : Analysis of algorithms	100.0%	-
Week 1 Quiz	-	100.0%
Week 2 : Searching and sorting	100.0%	-
Week 2 Quiz	-	100.0%
Week 2 Programming Assignment	-	100.0%
Week 3 : Graphs	100.0%	-
Week 3 Quiz	-	100.0%
Week 3 Programming Assignment	-	100.0%
Week 4 : Weighted graphs	100.0%	-
Week 4 Quiz	-	100.0%
Week 4 Programming Assignment	-	100.0%
Week 5: Data Structures: Union-Find and ...	100.0%	-
Week 5 : Divide and Conquer	100.0%	-
Week 5 Quiz	-	100.0%
Week 6: Data Structures: Search Trees	100.0%	-
Week 6: Greedy Algorithms	100.0%	-
Week 6 Quiz	-	100.0%
Week 6 Programming Assignment	-	100.0%

Week 7: Dynamic Programming	100.0%	-
Week 7 Quiz	-	100.0%
Week 7 Programming Assignment	-	100.0%
Week 8: Linear Programming and Networ...	100.0%	-
Week 8: Intractability	100.0%	-
Week 8 Quiz	-	100.0%
Text Transcripts	22.22%	-
Download Videos	100.0%	-
Lecture Material	100.0%	-
Problem Solving Session - Jan 2025	100.0%	-